

REVIEW

2024 Donald Seldin Lecture: New Insights Into Soluble (Pro)Renin Receptor in Hypertension and Kidney Disease

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ABSTRACT: Also known as ATP6AP2, the PRR ([pro]renin receptor) was originally cloned over 2 decades ago as a component of the renin-angiotensin system due to its property of nonproteolytic activation of prorenin. After heated debates, this concept has been confirmed and expanded with advances in sPRR (soluble PRR) research. sPRR is a 28-kDa protein that is derived from the extracellular domain of PRR and is primarily released by site-1 protease. Circulating sPRR is elevated in patients with multiple diseases, including diabetes, hypertension, preeclampsia, and kidney disease. A decade ago, sPRR was found to have an antidiuretic function that occurs through the modulation of vasopressin signaling in the collecting duct. This finding signified the beginning of a new chapter in understanding the biological function of sPRR in multiple organ systems, particularly the kidney, and blood pressure regulation. In the past few years, numerous groundbreaking discoveries in this field have shed new light on the complex biology of sPRR, including its heterogeneity due to proteolysis and the multifaceted interaction between sPRR and the intrarenal renin-angiotensin system. The goal of this review is to summarize new advances in this emerging field of sPRR research.

Key Words: biology ■ goals ■ kidney ■ pregnancy ■ renin

The renin-angiotensin system (RAS) plays an essential role in a wide variety of physiological and pathophysiological processes, ranging from homeostatic control of cardiovascular and renal functions to the pathogenesis of major human diseases such as chronic kidney disease (CKD), cardiovascular disease, and several types of hypertension.¹ To date, the anti-RAS regimen represents the cornerstone therapy for hypertension and CKD, as highlighted by the wide use of angiotensin-converting enzyme inhibitors and angiotensin receptor blockers.² Behind its pleiotropic actions in multiple systems, the RAS has evolved into 2 distinctive systems, for example, the blood-borne systemic RAS involving interaction of RAS components generated from different tissues, and the local RAS operating largely within an individual organ.^{3–5} The local RAS in the kidney is often referred to as the intrarenal RAS. A large body of experimental evidence has established a direct role of the intrarenal RAS in promoting renal fibrosis and inflammation

in various animal models of kidney disease.^{6,7} In parallel, the local RAS in the placenta has been implicated in the development of preeclampsia.⁸ In recent years, our understanding of the local system has been facilitated by the discovery of the PRR ([pro]renin receptor), also known as ATP6ap2, particularly the sPRR (soluble PRR), as a key regulator of the local RAS.^{9–11}

Nguyen et al⁹ cloned a full-length PRR in 2002, which was 4 years after the discovery of M8.9, the intracellular fragment of this protein, a subunit of H⁺-ATPase.¹² PRR was originally described as a nonproteolytic activator of prorenin and renin and a transducer of signal transduction.⁹ PRR has now emerged as a multifunctional protein that is implicated in a wide variety of developmental and physiopathological processes in different organs.^{5,9,13} At the structural level, PRR is composed of a large extracellular domain, a transmembrane domain, and a short intracellular domain. Following the cleavage, PRR exists in 3 distinctive isoforms: PRR (the full-length protein),

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Nonstandard Abbreviations and Acronyms

2K1C	2-kidney, 1-clip
AA	autoantibody
ADAM19	metalloproteinase 19
Ang II	angiotensin II
AT1R	angiotensin type 1 receptor
CKD	chronic kidney disease
IC	intercalated cell
PRR	(pro)renin receptor
RAS	renin-angiotensin system
S1P	site-1 protease
sPRR	soluble (pro)renin receptor

sPRR (the extracellular domain), and M8.9 (the intracellular domain), forming the basis of the functional diversity of this protein.^{14,15}

Figure 1 shows a timeline of research on sPRR and highlights some of the major events in the field. In 2006, the existence of sPRR was conceptualized based on the identification of a potential cleavage site in PRR.¹⁵ Three years later, the production and release of sPRR through furin-dependent cleavage were experimentally demonstrated,¹⁰ which was immediately followed by the development of an ELISA method for the measurement of sPRR in biological fluids.^{16,17} This assay has supported numerous clinical studies, which have revealed that serum levels of sPRR are increased during early pregnancy,¹⁸ preeclampsia,^{19,20} gestational diabetes,^{18,21} heart failure in patients with renal dysfunction,²² obstructive sleep apnea syndrome,²³ and CKD due to hypertension and type 2 diabetes.²⁴ The clinical studies support the idea that sPRR may serve as a biomarker for predicting outcomes and severity of multiple human diseases. In 2016, sPRR was reported to have a biological function in the regulation of renal water reabsorption.²⁵ In 2017, 2 independent groups reported that S1P (site-1 protease) is the predominant source of sPRR^{26,27} following previous reports on furin¹⁰ and ADAM19²⁸ as enzymes involved in PRR cleavage. In 2021, sPRR was found to directly interact with and activate AT1R (angiotensin type 1 receptor) in vascular endothelial cells, which led to endothelial dysfunction and obesity-related hypertension.²⁹ Within the kidney,

PRR is predominantly expressed in the distal nephron, particularly the collecting duct, where sPRR plays a major role in the regulation of electrolyte and fluid homeostasis and blood pressure, as well as in pathological processes, largely through its interaction with the intrarenal RAS.^{11,30,31} The current review will examine recent literature concerning new advances in the PRR cleavage processes and their functional implications, and the complex relationship between sPRR and the local RAS in the context of hypertension and kidney disease.

Heterogeneity of sPRR and Functional Implications

In 2009, Cousin et al¹⁰ identified furin as the first PRR-cleaving enzyme, which catalyzes the production of sPRR in the trans-Golgi network of LoVo colon cancer cells by recognizing a consensus sequence of R²⁷⁵-K-T-R²⁷⁸. Furin belongs to the proprotein convertase superfamily involved in the activation of functionally important protein precursors.³² LoVo colon carcinoma cells that are devoid of active furin are unable to secrete sPRR. In Chinese hamster ovary cells, sPRR production was inhibited after transfection with a plasmid coding for the α 1-antitrypsin Portland variant, an inhibitor of furin, or by mutagenesis of the furin cleavage site. However, this study is limited in that furin-dependent release of sPRR was demonstrated using artificial systems involving cell lines and transfection with plasmids.

In 2011, Yoshigawa et al²⁸ reported disintegrin and ADAM19 (metalloproteinase 19) as the second sPRR-generating enzyme after furin. They found that transfection of ADAM19 evoked the shedding of PRR, while transfection of dominant-negative ADAM19 suppressed it.²⁸ However, no consensus site for ADAM19 was identified in PRR. Given the controversial reports concerning furin versus ADAM19, it seemed imperative to identify the true cleavage enzyme responsible for sPRR production.

The subsequent discovery of S1P as the predominant cleavage enzyme represents a milestone in studying the PRR cleavage process. In 2017, Nakagawa et al²⁷ and our group²⁶ independently identified S1P as an essential enzyme for the release of sPRR in CHO cells expressing human PRR and HK2 cells loaded with bovine albumin,

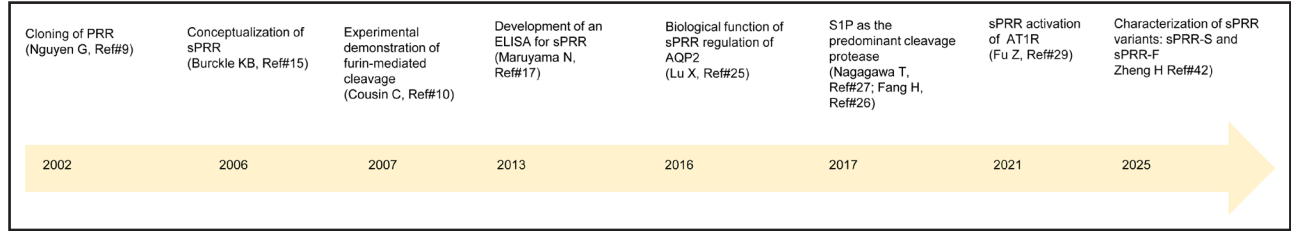


Figure 1. Timeline of characterization of sPRR (soluble [pro]renin receptor) during the past 2 decades.

respectively. Inhibition of S1P in HK2 cells reduced albumin-induced sPRR in the media by over 90%, suggesting that S1P is the predominant source of sPRR. Importantly, a large number of *in vivo* studies from our group and others support S1P-derived sPRR as an important player in various physiological and pathophysiological processes, including the physiological control of Na⁺ and water balance,^{33,34} the development of various types of hypertension induced by Ang II (angiotensin II),^{35,36} aldosterone-salt,³⁷ diet-induced obesity,²⁹ preeclampsia,^{38,39} and the pathogenesis of kidney diseases such as acute kidney injury⁴⁰ and CKD.^{6,41}

Besides the establishment of S1P as the predominant sPRR-releasing protease, advances have been made in understanding the distinctive role of furin in the PRR cleavage process. Furin inhibition does not affect the amount of sPRR released into the media, which rules out its involvement in the initial cleavage.²⁷ Interestingly, furin inhibition causes a size shift of sPRR in the media by ≈ 1 kDa, but not in the cell lysate, which indicates the possibility that furin may make a second cut in the released sPRR in the media.²⁷ This small size shift is consistent with the predicted cleavage site of furin, which removes 3 amino acids from the C-terminus of PRR. Consistent with this notion, we found that sPRR protein levels in mouse adipose tissue were slightly higher than those in the liver and kidneys.³⁶ The *in vitro* and *in vivo* evidence suggests the existence of 2 isoforms of sPRR: the intact sPRR generated by a single S1P cleavage (sPRR-S1P) and a new variant of sPRR following the second cut by furin (sPRR-furin), as described in our recent study.⁴² For simplicity, we would like to name sPRR-S1P as sPRR-S, and sPRR-furin as sPRR-F.

We functionally characterized the 2 sPRR variants during epithelial sodium channel (ENaC) regulation.⁴² In this study, the patch-clamp technique was used to measure ENaC open-channel probability in A6 cells exposed to overexpressed sPRR-S and sPRR-F. Surprisingly, ENaC open-channel probability was rapidly increased (within minutes) by sPRR-F but was unaffected by sPRR-S. Consistent with this observation, sPRR-F chronically stimulated α -ENaC expression (within hours) without affecting the β or γ subunit, but sPRR-S showed no effect. These results demonstrate the distinctive properties of the 2 sPRR variants in ENaC regulation. More importantly, this study marks the beginning of a new area of research concerning the functional roles and mechanisms of the 2 sPRR variants in various physiological and pathophysiological processes. Many issues still need to be addressed in future studies. How do the last 3 amino acids at the C-terminus determine the selectivity of binding of sPRR to its receptor(s), such as AT1R? How can we obtain *in vivo* evidence to specifically address the functional role of sPRR-F? Questions also remain about whether furin modification is necessary for the broad actions of sPRR in the kidney or the cardiovascular system other than ENaC regulation.

sPRR Regulation of Energy Metabolism

Emerging evidence suggests a potential role of sPRR in the regulation of energy metabolism. In a mouse model of lithium-induced polyuria, administration of a histidine-tagged sPRR (sPRR-His) did not affect the symptoms of nephrogenic diabetes insipidus.⁴³ It is surprising, however, that it induced a white-to-brown fat transition, known as the beiging phenomenon, which provided an early clue about the metabolic action of sPRR.⁴³ Subsequently, the recombinant sPRR exhibited therapeutic potential by lowering body weight, increasing energy expenditure, and improving insulin sensitivity, in DIO mice.⁴⁴ In further support of the metabolic action of adipose-derived sPRR, PRR has been identified as the target gene of PPAR γ in adipocytes. Inducible adipocyte-specific deletion of PPAR γ , a master regulator of energy metabolism, reduced circulating sPRR by $\approx 70\%$.⁴⁴ Together, the existing evidence seems to suggest the existence of 2 functional sPRR variants: the protective adipose-derived sPRR-S and the renal-derived pathogenic sPRR-F. The distinctive roles of the 2 sPRR variants are illustrated in Figure 2.

A recent preprint article reports a sex-specific phenotype in obesity-related metabolic syndrome and vascular dysfunction induced by adipocyte-specific overexpression of human sPRR.⁴⁵ Improvement in energy metabolism and vascular function was observed in female obese mice, but not in male mice.⁴⁵ However, a caveat of this mouse model is that circulating sPRR remained unchanged following adipocyte-specific overexpression of human sPRR.

sPRR Signaling via AT1R in Preeclampsia and Other Conditions

sPRR has been implicated as both a biomarker and a pathological mediator in preeclampsia.^{46,47} PRR is abundantly expressed in the placenta.⁴⁸ The source and cleavage pathway of sPRR during pregnancy have been investigated by several groups. Morosin et al⁴⁹ reported that syncytiotrophoblasts, the major placental secretory layer in direct contact with maternal blood, express PRR and secrete sPRR. However, inhibition of either furin (siRNA) or S1P (PF-429242) in term primary trophoblast cells did not affect sPRR production. In contrast, decanoyl-RVKR-CMK, a broad subtilisin/Kex2p-like proprotein convertase inhibitor that blocks the activity of all 7 convertases (PC1, PC2, PC4, PACE4, PC5, PC7, and furin), effectively reduced sPRR production. These results suggest the existence of a yet unidentified protease involved in the PRR cleavage pathway in the placenta. However, Suda et al⁴⁸ showed that in the human placental trophoblasts, the precursor of syncytiotrophoblasts, sPRR was induced by hypoxia, which was significantly blunted by inhibition of either furin or S1P.

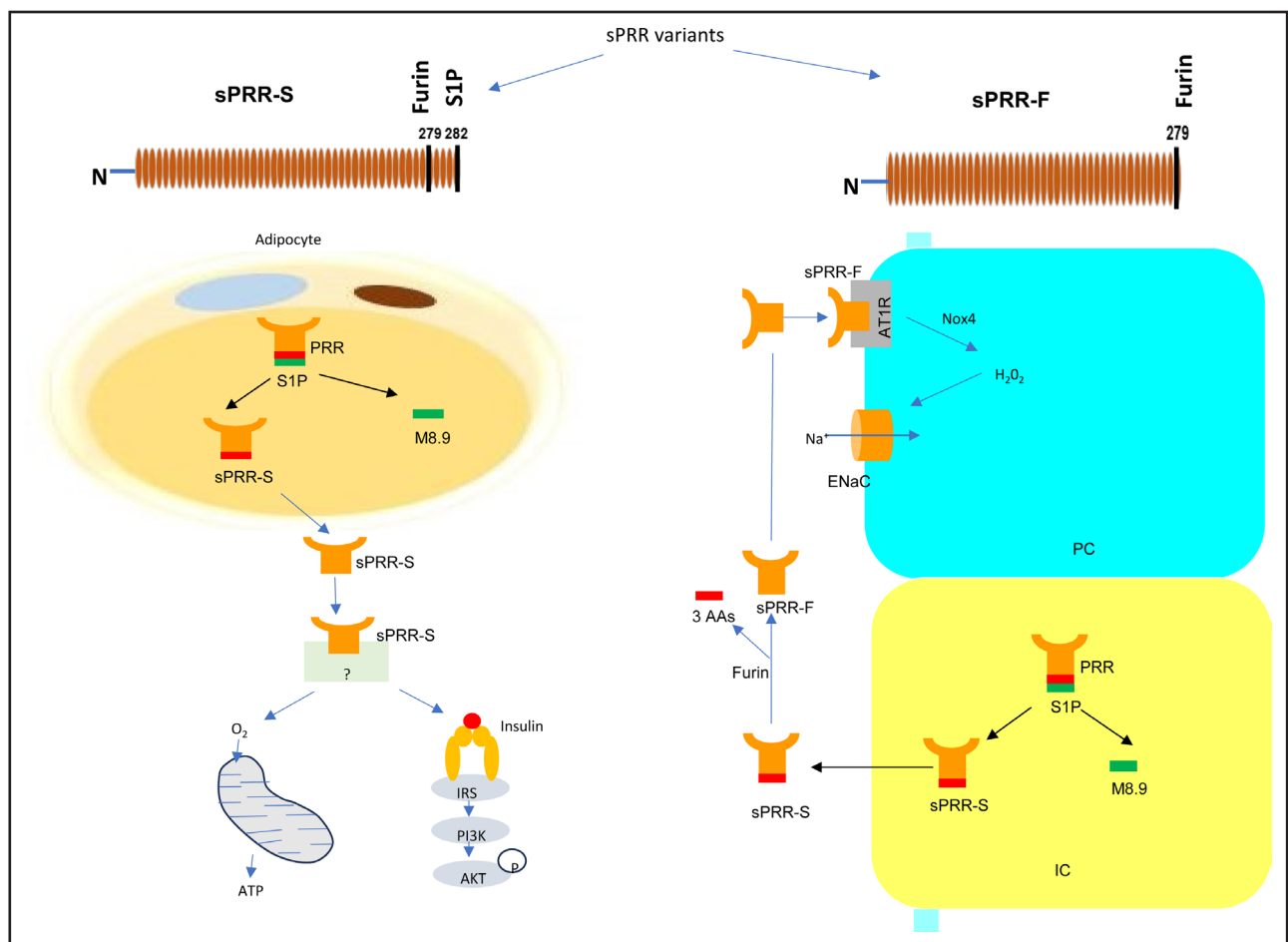


Figure 2. The proposed roles and mechanisms of sPRR (soluble [pro]renin receptor) variants.

sPRR exists in 2 distinctive variants, sPRR-S produced mainly in the adipocyte via a single cut by S1P (site-1 protease) and sPRR-F generated by furin-mediated removal of the last 3 amino acids at the C-terminus of sPRR-S. It is proposed that within the distal nephron sPRR-S first released by S1P remains inactive and upon furin-mediated removal of the last 3 amino acids at the C-terminus, inactive sPRR-S is converted to active sPRR-F. sPRR-F is capable of activating epithelial sodium channel (ENaC) activity in the principal cell (PC). It is further hypothesized that sPRR-S is primarily generated by the adipocyte to contribute to the known functions of adipose sPRR in sensitizing insulin action and promoting energy expenditure. IC indicated intercalated cell.

The reason for the discrepancy between the 2 studies is unclear but could be related to differences in the cell models or experimental conditions, including the oxygen level (normoxia versus hypoxia) and the differentiation status (syncytiotrophoblasts versus trophoblasts).

During the investigation of the vascular action of sPRR, we found that exposure of human umbilical vein endothelial cells to sPRR activated NF- κ B, downregulated phosphorylation of eNOS (endothelial nitric oxide synthase), and induced apoptosis, similar to the effects of AT1R activation.²⁹ This phenomenon occurred in the absence of increased Ang II in the media and resisted to angiotensin-converting enzyme inhibition, which suggested that sPRR might act via a signal transduction pathway independently of renin activity. Although Ang II is a well-known ligand of AT1R, this receptor can also be activated by Ang II-independent factors such as mechanical stress in the heart⁵⁰ and AT1 autoantibodies (AT1-AA) during preeclampsia.^{51,52} Therefore, we examined

the possibility that sPRR may act via AT1R to elicit a proinflammatory response in cultured human umbilical vein endothelial cells. Inhibition of AT1R using losartan or siRNA effectively reversed the proinflammatory and proapoptotic effects of sPRR in these cells.²⁹ Immunoprecipitation coupled with autoradiography experiments and radioactive ligand competitive binding assays indicated that sPRR directly interacts with AT1R via Lys199 and Asn295, which are the 2 residues that are critically important for the binding to Ang II and losartan. sPRR-His increases blood pressure in obese C57BL/6 mice, which was reversed by concurrent treatment with losartan, but not captopril. These results support the importance of the direct AT1R activation, rather than increased renin activity, in mediating prohypertensive action of sPRR in obesity. In agreement with this notion, Gatineau et al⁵³ showed that sPRR infusion results in increased systolic blood pressure in male mice that have been fed a high-fat diet.

Preeclampsia involves persistent high blood pressure that occurs between the 24th and 27th weeks of pregnancy or soon after giving birth (postpartum preeclampsia).⁵⁴ The pathophysiology of this disease features systemic vascular endothelial damage.⁵⁵ The cause of this disease is multifactorial, involving the activation of AT₁R by AT₁-AA,⁵⁶ the release of the VEGF receptor antagonist soluble sFlt-1 (fms-like tyrosine kinase-1),⁵⁷ and recently the overproduction of sPRR.^{39,58,59} Numerous studies have examined the potential role of the RAS in preeclampsia, revealing a complex role of this system. It has been well documented that circulating renin and Ang II levels are suppressed in preeclampsia,^{60,61} arguing against the involvement of the systemic RAS. Interestingly, Ang II signaling via AT₁R is decreased in normal pregnancy but enhanced in preeclampsia.^{62,63} These results suggest an Ang II-independent action of AT₁R, likely within a local RAS during preeclampsia.

To date, the enhanced AT₁R signaling in preeclampsia has largely been attributed to AT₁-AA, which are increased in patients with preeclampsia and are capable of directly binding and activating AT₁R, resulting in vascular endothelial dysfunction.⁶⁴ Recently, Herrock et al¹⁵⁶ reported that when administered to normal pregnant rats, placental CD19+B cells isolated from patients with preeclampsia or a history of COVID-19 infection released AT₁-AA, inducing hypertension and inflammation. Overall, based on the compelling evidence in the field, a consensus can be reached that AT₁-AA not only serve as a biomarker but also as a causal factor in the pathogenesis of preeclampsia through the direct activation of AT₁R.

Besides AT₁-AA, a significant number of recent studies suggest a potential role of sPRR as an alternative activating agonist of AT₁R during preeclampsia. Clinical studies report that compared with normotensive pregnancies, circulating sPRR is increased in patients with preeclampsia²⁰ or in those with pregnancy complications, such as fetal growth restriction⁶⁵ or gestational diabetes.⁶⁶ While increased plasma sPRR has been considered a biomarker of preeclampsia, a series of studies have defined a causal role of the sPRR/AT₁R axis in the pathogenesis of preeclampsia.^{38,39} In this regard, the exposure of human uterine microvascular endothelial cells to a recombinant sPRR induced increases in the mRNA expression of endothelial dysfunction markers, endothelin-1, VCAM-1, and ICAM-1, which were all blunted by losartan and an AT₁R siRNA. The administration of sPRR to pregnant rats induced a preeclamptic phenotype, including increased blood pressure and endothelial dysfunction.³⁸ Furthermore, docking analysis indicated that AT₁R residues Lys199, Tyr184, and Thr260 as well as sPRR residues Asn292 and His256 are crucial for sPRR-AT₁R complex formation.³⁹ The accuracy of the docking analysis is reflected by the identification of Lys199 in AT₁R, although it did not identify another key residue, Asn295. Overall, the

docking results provide new insight into the interaction between sPRR and AT₁R, but the precise structure of the complex needs to be resolved by performing physical structural analysis using cryogenic electron microscopy (cryo-EM) or X-ray crystallography. Both AT₁-AA and sPRR bind and activate AT₁R, contributing to the pathogenesis of preeclampsia. Whether there is a direct interaction between AT₁-AA and sPRR remains elusive. Furthermore, the relationship between sPRR and sFlt-1, another well-established mediator of preeclampsia, remains unknown.⁶⁷ It seems conceivable that AT₁-AA and sPRR may cooperate in activating AT₁R, leading to upregulation of sFlt-1, ultimately causing preeclampsia pathophysiology.^{68–70} Given the well-described developmental role of the RAS in the kidney,⁷¹ RAS inhibitors may impair fetal kidney development and thus are contraindicated in the management of preeclampsia. Rather than broadly inhibiting the RAS with existing RAS inhibitors, targeting AT₁-AA and sPRR may specifically inhibit preeclampsia-related activation of AT₁R, offering a novel intervention for this devastating disease, and may also be devoid of the toxicity of renal developmental abnormalities. Indeed, inactivation of multiple RAS components including angiotensinogen, angiotensin-converting enzyme, or AT₁R induces an identical phenotype of renal developmental abnormalities, suggesting involvement of an intact RAS in nephrogenesis,^{72,73} whereas preeclampsia is associated with overactivation of AT₁R by AT₁-AA and sPRR but suppressed maternal circulating levels of renin and Ang II.^{62,63,74}

sPRR Regulation of Intrarenal Renin

Within the distal nephrons, PRR and renin expression are restricted to the intercalated cells (ICs)⁷⁵ and principal cells,⁷⁶ respectively, which provides physical evidence that favors their interaction. A decade ago, we proposed an intriguing mechanism of action of sPRR in the CD: sPRR, generated and released by the ICs, would act in a paracrine manner to transcriptionally activate AQP2 (aquaporin-2) and α -ENaC in the principal cells via β -catenin signaling.²⁵ In addition, increasing evidence supports complex relationships between sPRR and renin in the distal nephron. Although the concept of sPRR mediation of the nonproteolytic activation of prorenin has been widely accepted, we found that the transcriptional mechanism plays a more important role. For example, the exposure of M-1 CD cells to sPRR-His resulted in upregulation of renin mRNA expression and protein content through β -catenin signaling, which cannot be explained by the conformational alteration of prorenin/renin.³⁶ An animal model of Ang II-induced hypertension showed suppressed systemic RAS but overactivated intrarenal RAS, as evidenced by activation of renin in the CD.⁷⁷ The Ang II-induced increases in urinary and renal medullary renin were blunted by CD-specific deletion of

PRR,⁷⁷ pharmacological inhibition of S1P,³⁵ and mutagenesis of the cleavage site of PRR.³⁶ The 2-kidney, 1-clip (2K1C) Goldblatt model⁷⁸ is more clinically relevant to human renovascular hypertension than the animal model of chronic infusion of pharmacological doses of Ang II. Systemic RAS activation due to increased renin release from the juxtaglomerular cells (JG cells) is well described in renovascular hypertension,⁷⁹ but increasing evidence supports the involvement of intrarenal RAS.⁸⁰ We recently found that CD-specific deletion of PRR remarkably attenuated renal fibrosis and inflammation associated with a rather modest attenuation of hypertension.⁷ The robust phenotype in renal pathology cannot be explained by a modest change in blood pressure. sPRR production was attenuated in clipped null mice in parallel with suppressed intrarenal renin levels. Along this line, a significant reduction of blood pressure by sodium nitroprusside did not lead to any improvement in renal injury in this model. These results may suggest a predominant role of CD PRR/sPRR in mediating renal fibrosis via activation of the intrarenal RAS. It seems possible that the intrarenal RAS only represents part of the redundant mechanisms responsible for the development of hypertension in the 2K1C model. Likely, the cooperation of the systemic RAS and the intrarenal RAS may be more important for the hypertension phenotype.

A direct role of CD PRR/sPRR and intrarenal RAS in renal fibrosis has recently been tested by using a mouse model of unilateral ureteral obstruction, a widely used model of renal fibrosis.⁶ In this study, obstruction-induced renal fibrosis was consistently attenuated by CD-specific deletion of PRR or renin, or by administration of the S1P inhibitor PF-429242, associated with suppressed intrarenal renin levels.⁶ In cultured renal epithelial cells, sPRR-His induced fibrosis, which was blunted by losartan, indicating involvement of AT1R in addition to renin.⁴¹ These results support the idea that PRR/sPRR-dependent stimulation of CD renin expression together with the direct activation of AT1R induces the intrarenal RAS, leading to obstruction-induced renal fibrosis.

The experimental approach with the administration of PF-429242 or mutagenesis of the PRR cleavage site used in the above-described studies is limited in that the systemic inhibition of sPRR production may not directly address the function of IC-derived sPRR. Future studies are needed to examine whether S1P is localized to the ICs. If so, IC-specific deletion of S1P can be generated using the Cre-loxP strategy to definitively assess the role of IC-derived sPRR in various models of kidney disease and hypertension.

Therapeutic Potential of PRO20

PRO20 is a well-documented PRR decoy peptide that is capable of blocking PRR-dependent signal transduction, lowering blood pressure, and protecting against

renal injury.^{60,81} These effects have been observed in a variety of animal models of hypertension and kidney disease, including albumin-overloading nephropathy,²⁶ cyclosporine A nephropathy,⁸² hypertension induced by deoxycorticosterone-salt⁶⁰ and Ang II,⁸¹ and 5/6 nephrectomy.⁶¹ In most of these cases, PRO20 acted via inhibition of the intrarenal RAS. The compelling preclinical evidence from these studies suggests a translational potential of PRO20 in inhibiting the intrarenal RAS to treat various types of hypertension and kidney diseases. To date, no obvious toxicity has been noticed for this peptide. Importantly, PRO20 exhibited therapeutic potential in preeclampsia, for which an effective and safe therapy is still missing. In this regard, PRO20 treatment of human uterine microvascular endothelial cells mitigated sPRR- and preeclamptic serum-induced expression of endothelial dysfunction markers endothelin-1, VCAM-1, and ICAM-1.³⁹ Furthermore, docking analysis revealed a structural basis for PRO20-mediated blockade of sPRR-AT1R complex formation. The in vitro evidence provides proof of concept for the use of PRO20 to treat preeclampsia. While the in vitro data are encouraging, it is imperative to examine the therapeutic potential as well as a safety profile of this peptide in clinically relevant animal models of preeclampsia, such as those induced by placental ischemia and infusion of preeclampsia-related factors in pregnant animals.⁸³

Although the concept of the sPRR-AT1R axis has been well demonstrated in the vasculature, it remains largely unexplored whether this mechanism operates in the setting of renal control of Na⁺ and water balance and kidney disease. sPRR is known to rapidly increase ENaC open-channel probability, which likely occurs via Nox-4-dependent generation of H₂O₂³⁴ and chronically stimulates transcription of its target genes, such as α -ENaC³⁴ and AQP2,²⁵ as well as renin,³⁶ via β -catenin signaling. It seems reasonable to speculate that AT1R may mediate some or all of the acute and chronic actions of sPRR, which need to be tested in future studies. We have shown that sPRR directly binds the membrane protein Frizzled-8, which leads to activation of β -catenin signaling in CD cells.²⁵ In support of this notion, the β -catenin pathway mediates the AT1R-dependent profibrotic response to Ang II in M-1 CD cells.⁸⁴ We suspect that binding to AT1R may be sufficient to confer the acute action of sPRR such as the activation of ENaC activity, whereas a cooperation between AT1R and Frizzled-8 may be necessary for sPRR-mediated transcriptional activation of the target genes.

sPRR Regulation of Intrarenal Generation of Aldosterone

CYP11B2 catalyzes the last unique step in aldosterone biosynthesis implicated in a wide range of physiological

and pathophysiological processes. Although the adrenal zona glomerulosa has been considered as the predominant source of aldosterone, local generation of aldosterone has been reported before in the brain, the cardiovascular system, adipose tissue, and the kidneys.^{85,86} However, skepticism exists that the extra-adrenal tissues may not produce sufficient aldosterone to mediate pathophysiological roles.

Two early studies from the Helmy Siragy group observed intrarenal aldosterone biosynthesis in experimental animals with diabetic kidney disease.⁸⁷ Following adrenalectomy, plasma aldosterone was reduced to an undetectable level,⁸³ which indicated that the adrenal glands were the sole source of circulating aldosterone. In contrast, intrarenal aldosterone in adrenalectomized rats only exhibited a $\approx 25\%$ reduction compared with sham controls. Furthermore, renal expression of CYP11B2 was demonstrated at both the mRNA and protein levels and localized to a number of renal structures, including glomeruli and proximal tubules.⁸⁸ Renal expression of CYP11B2 is increased by diabetes, and its inhibition by the CYP11B2 inhibitor FAD286 improved CKD via repression of renal inflammation and fibrosis.⁸⁷ Moreover, a series of recent studies from our group showed that renal CYP11B2 expression was under the control of PRR and COX-2 and is implicated in the regulation of Na⁺ and K⁺ homeostasis.^{89–91} The new evidence has sparked interest in pursuing the potential role of intrarenal CYP11B2 in hypertension and kidney disease.

A preprint article from our group recently presented preliminary evidence for the pathological role of intrarenal aldosterone in a mouse model of a role of intrarenal aldosterone biosynthesis catalyzed by CYP11B2 in a mouse model of 2K1C-induced hypertension.⁷ In this study, the Pax8-rtTA-LC1-Cre was used to produce tetracycline-inducible deletion of CYP11B2 in the renal tubules, and the phenotype of hypertension and renal fibrosis was examined in the 2K1C model. Null mice exhibited a

partial blockade of the hypertensive response but a much greater improvement in renal fibrosis and inflammation at 4 weeks post-2K1C. The phenotype is almost analogous to that found in mice with CD-specific deletion of PRR or renin.⁷ Together, these results seem to suggest the existence of an intact intrarenal RAAS under the control of CD PRR/sPRR.

SUMMARY AND PERSPECTIVE

The past decade has witnessed significant advancements in understanding the source, the cleavage process, and biological functions of sPRR, particularly in the kidney. Among multiple potential cleavage proteases, S1P has been established as a predominant source of sPRR. Recent evidence suggests that sPRR exists in 2 different variants, sPRR-S released by S1P and sPRR-F produced by furin-dependent removal of the last 3 amino acids at the C-terminus of sPRR-S. Among a limited number of mouse tissue types examined, sPRR-S protein was only detected in adipose tissue, whereas sPRR-F protein was found in other tissues, including the kidney. Functional studies showed that sPRR-F stimulated ENaC open-channel probability, whereas sPRR-S was without an effect. Besides ENaC regulation, sPRR-F may contribute to other known actions of sPRR in the kidney and the vasculature, such as vasopressin signaling, intrarenal renin expression, renal fibrosis and inflammation, hypertension, etc. It is known that PPAR γ -dependent adipocyte-derived sPRR represents an important player in the regulation of energy metabolism and insulin sensitivity. To what extent sPRR-S mediates the metabolic action of sPRR needs to be investigated in future studies. If so, it will be interesting to examine the underlying mechanism by which sPRR-S controls cellular energy metabolism.

The concept of sPRR as a regulator of tissue RAS has been well established. It was originally reported that binding of PRR or sPRR to prorenin induced

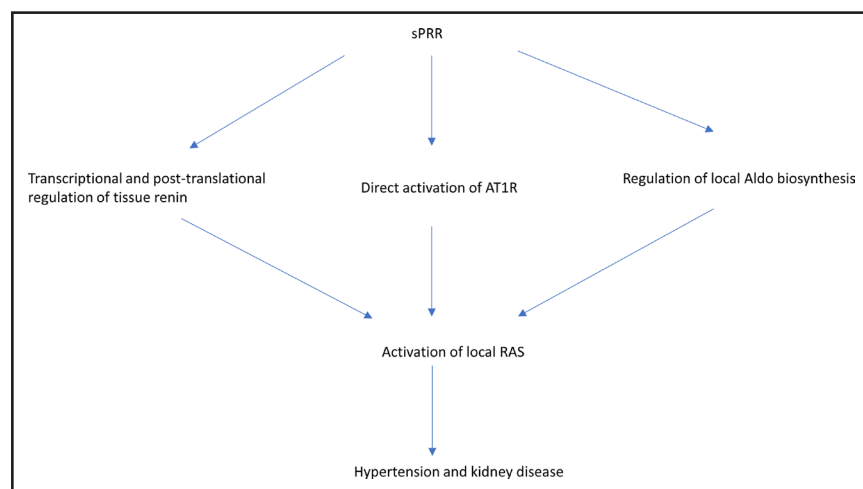


Figure 3. sPRR (soluble [pro]renin receptor) regulation of multiple components of the local renin-angiotensin system (RAS).

First, sPRR can induce tissue renin activity through conformational activation of tissue prorenin and renin. Apart from renin activity regulation, sPRR activates transcription of tissue renin via β -catenin signaling. Second, sPRR directly binds and activates AT1R (angiotensin type 1 receptor) independently of Ang II (angiotensin II) generation. Finally, preliminary evidence indicates that sPRR upregulates intrarenal aldosterone (Aldo) generation. Therefore, sPRR likely functions as a key regulator of the local RAS through modulation of multiple components of this system.

nonproteolytic activation of prorenin due to the conformational changes. However, increasing evidence suggests that sPRR activates CD renin primarily through transcriptional upregulation of renin mRNA via β -catenin signaling. In addition, sPRR directly binds to and activates AT1R to elicit proinflammatory, proapoptotic, and profibrotic responses, contributing to obesity- and preeclampsia-related endothelial dysfunction, hypertension, and CKD. More recently, PRR/sPRR were shown to target intrarenal aldosterone generation during clipping-induced renal fibrosis and hypertension. Together, these results suggest that sPRR may serve as a key regulator of local RAS in multiple tissues by targeting multiple RAS components (Figure 3). A better understanding of sPRR-dependent regulation of the local RAS will shed light on the mechanisms of multiple physiological and pathophysiological processes and facilitate the development of novel therapeutic targets for the management of major human diseases such as CKD, cardiovascular disease, and preeclampsia.

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Disclosures

None.

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